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Games, but no fun

Mercury Computer faces challenge as techies embrace PlayStation 3

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Three years after striking a deal to become one of the first companies to use the powerful PlayStation 3 chip outside of the gaming industry, Mercury Computer Systems Inc. is struggling to get past the first level.

Instead of racking up points and profits, Chelmsford-based Mercury is a turnaround story.

Its head start with the Cell Broadband Engine processor, which helps make video games an addictive experience, has translated into adoption by some key customers, such as the U.S. military, but not sustained revenue growth.

Meanwhile, scientists and academics have discovered that lashing together several or dozens of $499 PlayStation 3s together can create a cheap supercomputer.

Not to be outdone, Mercury has a $399 software development kit that uses Terra Soft Solutions Inc.'s Yellow Dog Linux operating system that converts a PlayStation into a supercomputing node. It also has worked with IBM Corp. to build cell versions of Big Blue's popular blade server, which costs thousands of dollars.

The processing guts of a PlayStation 3, the product of a partnership among IBM, Sony Corp. and Toshiba, have more in common with high-end supercomputers than they do with the toys they share shelf space with at stores, said Kenneth Roe, who studied the PlayStation 3's cell processor as part of his computer science master's degree project at the University of New Haven.

Gaurav Khanna, an astrophysicist at the University of Massachusetts-Dartmouth, said he combined PlayStation 3 consoles to see if they could do the heavy computing that he would normally do with rented time on supercomputers at universities across the country.

"It's a very powerful beast and you can run your own programs on it," he said, adding that he doesn't have to wait several weeks to do an experiment on a university supercomputer.

He first put together eight PlayStation 3s last fall, some donated by Sony. Today, he has 16 machines daisy-chained together. The cost savings: a factor of 10, according to Khanna.

“I would have never been able to do this project on my own,” he said.

Khanna said he has fielded calls from scientists around the globe. They picked his brain on how they can build their own PlayStation-powered supercomputers.

Mercury Computer executives said they don’t see those initiatives as competition, but as seed projects that could spur further adoption — and demand — for the company’s cell processor technology.

“The PlayStation is a good tool for application development,” said Randy Dean, a vice president of integrated solutions at Mercury Computer. “When the PS3 first came out, we had our co-ops stand in line and we bought a whole bunch of them.”

Still, some initiatives around the country with the PlayStation 3 are creeping toward Mercury Computer’s turf.

And Mercury Computer needs a catalyst, not more competition.

The company reported fiscal 2008 revenue of $209.9 million — 3 percent year-over-year slip — and a net loss from continuing operations of $35.4 million. That was a slight improvement over the $42.8 million loss booked in the previous year.

In a deal with IBM in 2005 to use the cell processor, Mercury (Nasdaq: MRCY) said it would build new breakthrough computer systems for data-intensive applications. The company’s single largest product investment in fiscal years 2006 and 2007 was for microprocessor boards and system level products based on the cell processor and related software, according to its financial statements.

“We’ve been pushing the cell for three years now,” Dean said “We’ve sold thousands of cell processors for deployment and evaluation. ... There’s been quite a lot of adoption of the technology.”

The challenge of the cell processor is creating applications that work for radar, digital X-rays, seismic processing and telecommunications. Better medical imaging through better computing power could lead to the earlier detection of disease “and potentially saving lives,” Mercury said in its June 28, 2005, press release heralding the deal with IBM.

Stanford University scientists have experimented with complex protein modeling and simulation by distributing the work over a network of processors. Some 39,000 PlayStations are involved. Mercury executives see projects similar to the one at Stanford as a favorable trend.

Massive computing is being deployed to replace trial and error. As experimentation gives way to simulation, Mercury hopes to step in with a system to do advanced computing.

And the cell chip used by Mercury is more powerful than what’s found inside a PlayStation 3, whose memory and bandwidth is limited compared to a Mercury setup.

Still, Mercury’s core customer, the U.S. military, seems fascinated by the PlayStation 3’s computing power. In February, for example, the U.S. Air Force Research Laboratory issued a request for proposal to buy 300 PlayStation 3s as part of a technology assessment of cell processors. Air Force officials declined to comment.

When asked about that project, Mercury Computer's Dean laughed.

"Ask me that question in six months," he said.

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